Weixuan Yuan

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Research interests: LLM reasoning and post-training; AI for mathematics and theoretical computer science.

Education & Research Positions

ETH Zürich — Ph.D. in Computer Science May 2025–Present
Advisor: Prof. Rasmus Kyng.

ETH Zürich — Research Assistant October 2024–April 2025
Rasmus Kyng's group.

ETH Zürich — M.Sc. in Data Science September 2022–October 2024
Willi Studer Prize (2025), awarded to the best graduate of the M.Sc. in Data Science program.

Xi'an Jiaotong University — B.Sc. in Mathematics September 2018–July 2022
Honors Youth Program

Selected Research

AuXSpec — Formalizing Programming Problems with LLMs

We study LLM-based translation of natural-language programming problems into formal specifications in Lean.

- Proposed using executable postconditions to test generated specifications efficiently; co-designed the Lean specification format.
- Co-developed a multistage LLM-based pipeline for data synthesis, verification, and repair. The team produced 3,037 formalized LeetCode problems and approximately 19.5K SFT samples.
- Independently reviewed 300 benchmark problems and audited 60 model-generated specifications.
- Contributed to discussions on post-training methods and reward design, and to analysis of results. After SFT/RL, the team's 27B model achieved SpecGen pass@4 of 60.3% on AuXSpec-Test and 84.7% on Verina.

Graph algorithms — FOCS 2025

Co-authored a paper giving the first deterministic almost-linear-time Gomory-Hu tree algorithm: $m^{1+o(1)}$ time for graphs with m edges. Presented the paper with Wuwei Yuan at FOCS 2025. [slides]

Selected Publications

Deterministic Almost-Linear-Time Gomory-Hu Trees
Amir Abboud, Rasmus Kyng, Jason Li, Debmalya Panigrahi, Maximilian Probst Gutenberg, Thatchaphol Saranurak, **Weixuan Yuan**, Wuwei Yuan.
IEEE Symposium on Foundations of Computer Science (FOCS 2025). [paper]

Optimal subsampling for principal component analysis
Xuehu Zhu[†], **Weixuan Yuan**[†], Zongben Xu, Wenlin Dai.
Science China Mathematics 68, 2993-3016, 2025. [paper]

Optimal subsampling for estimation of dimension reduction directions
Xinru Jia[†], **Weixuan Yuan**[†], Xingqiu Zhao, Xuehu Zhu.
Scandinavian Journal of Statistics 53(1), 575-611, 2026. [paper]

[†] Equal contribution. FOCS authors are listed alphabetically. Statistics work dates to my undergraduate studies (before 2023).

Under-Review Manuscript

AuXSpec: Learning to Autoformalize Program Specifications via Executable Postconditions
Co-first author; under review at ICLR 2027. Manuscript not yet publicly available.

Technical Skills

Programming: Python, Lean 4.
Theory: Algorithms; probability and statistics; optimization, including gradient-based methods.